KU Leuven

Faculty of Engineering Technology

Robotic Testbench for Collaborative Handling of Payloads with Varying Weights using Impedance Control, a Disturbance observer and Compliant Transmissions

---

Oskar De Clerck

A thesis submitted for the degree of
Master of Science in *industriële wetenschappen: Elektromechanica*

Academic Year 2025–2026

Promotor: ir. Maurizio Grossi
Copromotor: prof. dr. ir. Dino Accoto

Abstract

This thesis addresses [problem]. The main objective is to [objective].

We develop [methodology/approach] and demonstrate [key result]. The results show [findings].

Our main contributions are: (1) [first contribution], (2) [second contribution], and (3) [third contribution].

Keywords: keyword1, keyword2, keyword3, keyword4, keyword5

Acknowledgments

Contents

Abstract **i**

Acknowledgements **iii**

Nomenclature **xi**

1 Introduction **1**
1.1 Background and Motivation 1
1.2 Problem Statement 1
1.3 Thesis Outline 1

2 Literature Review **3**
2.1 Theoretical Background 3
2.1.1 Example subsection 3
2.1.2 Another subsection 3
2.2 State of the Art 3
2.3 Gap Analysis 3

3 System Design and Implementation **5**
3.1 System Overview and Requirements 5
3.2 Mechanical Design 5
3.3 Hardware Architecture 5
3.3.1 CAN Bus Topology 5
3.3.2 Sensor Architecture 5
3.4 Sensor and Actuator Modelling 5
3.4.1 SET Spring Model 5
3.4.2 Motor and Transmission Model 5
3.5 Control Architecture 5
3.5.1 State Machine 5
3.5.2 Gravity Compensation and Floating Mode 5
3.5.3 Weighing Mode 5
3.5.4 Carrying Mode 5
3.5.5 Put-Down Mode 5
3.6 Software Implementation 5
3.6.1 Real-Time Architecture 5

3.6.2 Configuration 6

4 Validation and Results **7**

4.1 Experimental Setup 7

4.1.1 Hardware Configuration 7

4.1.2 Parameter Identification 7

4.2 Spring and Sensor Characterisation 7

4.3 Controller Performance 7

4.4 Assisted Load Handling Evaluation 7

4.5 Discussion 7

5 Conclusion and Future Work **9**

5.1 Conclusion 9

5.2 Future Work 9

2D Manufacturing Drawings **11**

.1 CNC Part 1 11

.2 CNC Part 2 11

.3 CNC Part 3 11

Additional Experimental Results **13**

Source Code **15**

List of Figures

1.1 Caption text. 1

List of Tables

4.1 Example results table. 7

Nomenclature

Symbols

A	Cross-sectional area [m^2]
E	Young's modulus [Pa]
F	Force [N]
m	Mass [kg]
t	Time [s]
v	Velocity [m/s]
α	Angle [rad]
ω	Angular frequency [rad/s]

Abbreviations

AI	Artificial Intelligence
CAD	Computer-Aided Design
FEM	Finite Element Method
IoT	Internet of Things
PID	Proportional-Integral-Derivative

Chapter 1

Introduction

Assembly and packaging tasks often impose significant phyiscal strain on workers' joints, affecting both quality of life and productivity **greggiWorkRelatedMusculoskeletalDisorders2024**. This chapter provides context for this problem, explains its societal relevance and the objectives of this thesis.

1.1 Background and Motivation

Ergonomic challenges

Industrial assembly and packaging tasks remain among the most ergonomically demanding operations in manufacturing, requiring workers to perform repetitive and force–intensive motions. These activities which cause repetitive strain on different body parts can result in Musculoskeletal Disorders (MSDs). According to the World Health Organization, one in three people worldwide lives with a chronic MSD **gorassoHealthEconomicBurden2023**.

The European Agency for Safety and Health at Work considers MSDs to be the main occupational disease, affecting approximately 45 million workers within the European Union **gorassoHealthEconomicBurden2023**, causing billions of euros of lost revenue for companies and a great burden on the public health care system **ibvMusculoskeletalDisordersCaused2017**.

In 2018, 2.5 million Belgians were affected by one or more MSDs. Over a period of four years, from 2013 to 2017, these MSDs resulted in approximately 3 billion euros in medical expenses and 2 billion euros of lost revenue due to work absenteeism, corresponding to 0.3% of Belgian gross domestic product (GDP) in 2018 **gorassoHealthEconomicBurden2023**.

Reduced physical performance

Reduced work ability can be categorized in absenteeism and presenteeism. Absenteeism refers to the absence from work due to injuries sustained on the job. Presenteeism describes the situation in which workers continue to attend their expected working duties despite existing injuries, often due to concerns about job security. In such cases, work is performed in a self–protective manner to reduce physical strain. Both absenteeism and presenteeism result in reduced work efficiency, which leads to revenue losses and increased medical costs **oralEvaluationAbsenteeismPresenteeism2024**.

Physical human–robot interaction

Human–robot interaction can be classified into several levels based on the different types of interaction, as illustrated in Figure **??**. The robotic application in this thesis falls under human–robot collaboration: Human worker and robot interact with the same component simultaneously.

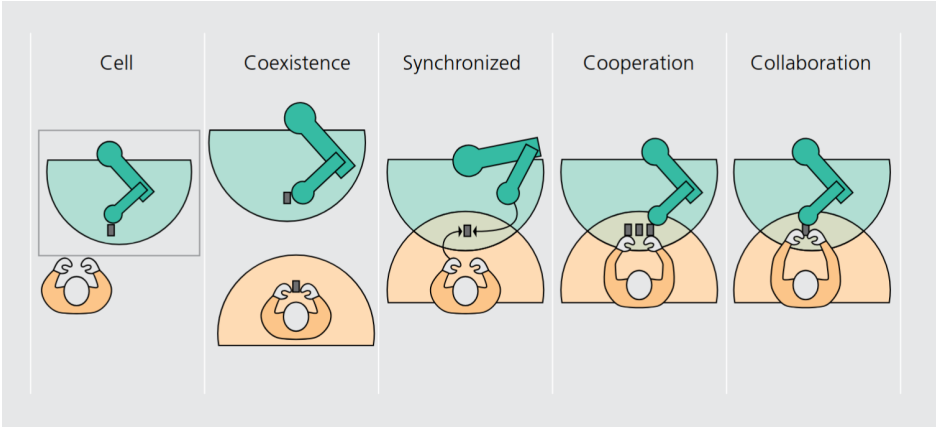

Figure 1.1: Levels of interaction between human and robot (reproduced from **benderLightweightRobotsManual2016**)

Robotic solutions

There is a clear need to address reduced work efficiency resulting from physical injuries since this improves the quality of life of an increasingly aging workforce, but also to reduce the impact on social welfare systems and increase overall economic productivity. Yet few collaborative applications exist in production facilities. Human and robots are either physically seperated, or work in a coexistence arrangement where no physical barrier seperates human and robot but no direct interaction takes place **benderLightweightRobotsManual2016**.

Implementations like this are only economically viable for large batch sizes, since a whole production line needs to be tuned towards that specific robotic solution **benderLightweightRobotsManual201** Cage–free robotic arms that share a workspace with a human worker are introduced to increase flexibility of the production line. Interactive robotic solutions for small batches and varying product sizes has become a clear need since 70% of collaboration between human worker and robotic solutions are found in assembly applications **benderLightweightRobotsManual2016**.

An industry survey reported that 86% of respondents were willing to invest in lightweight robotic solutions, indicating strong industrial interest. The primary motivations for introducing robotic arms in assembly environments are:

- Improving operational efficiency
- Improving ergonomics

Cage–free robotic solutions generally involve higher initial costs compared to traditional fenced systems due to safety certification requirements and CE conformity procedures. Com-

panies are however often willing to accept longer payback periods for implementing and testing company specific robotic solutions which increases ergonomics **benderLightweightRobotsManual2016**.

Limitations of existing collaborative robots

Traditional stiff robots in collaborative applications are heavily regulated to minimise the risk of injuries. These robots are lightweight, operate at low speeds and often lack mechanical bandwidth to react to unforeseen collisions in a safe and compliant way **benderLightweightRobotsManual2016s**

One solution for safe collaboration between human and robot are hand–guided robotic arms. Hand–guided robotic applications are defined as a direct physical interaction between human and robot, where the human operator has full control over the robot motion. In this application, a direct input device is located at or near the end–effector to measure input forces and simultaneously define the position of the human in the shared workspace. A direct consequence of this approach is that robotic applications with sensorised end–effectors can only react in a passive compliant manner when handled at a specific point on the robotic arm. Collisions or attemted handling of the robotic arm elsewhere will result in a safety stop of the robotic arm **sicilianoSpringerHandbookRobotics2016**.

1.2 Problem Statement

Existing robotic solutions that allow a human worker to hand–guide a robotic arm by grapsing it wherever is most convenient are, to the best of current knowledge, practically non-existent in assembly and packaging industries. When hand–guiding is discussed in literature, it is typically limited to sensorised end–effectors on conventional stiff robots **gregorHandGuidingVirtual2021hameedControlSystemDesign2023**. This absence in current industrial applications is further supported by surveys of end users, system integrators and researchers which states that collaborative robots are still primarily implemented as traditional stiff robots with limited or no direct human interaction **montiniCollaborativeRoboticsSurvey2024**. The proposed robotic solution must be useable without extensive training and intuïtive for operators without technical experience. To further enhance the return on investment, the system should also be portable, allowing the implementation of the robotic solution at different locations within an assembly or packaging facility wherever it is most needed.

Chapter 2

Literature Review

This chapter reviews the existing literature related to this research.

2.1 Theoretical Background

The fundamental theory underlying this work is...

- Impedance control
- Disturbance Observer
- Docker
- ROS2

2.1.1 Impedance control

2.1.2 Disturbance observer

2.1.3 Docker

2.1.4 ROS2

2.2 State of the Art

Recent advances in this field include...

2.3 Research Gap

While existing methods provide solutions for..., they have limitations:

- Limited scalability
- High computational cost
- Lack of robustness

This thesis addresses these gaps by...

Chapter 3

Methodology

This chapter describes the research methodology and proposed approach.

3.1 Overview

Figure **??** illustrates the overall methodology.

Make figure with 3D design, code, workspace (how it would be implemented in for example TroPack)

Figure 3.1: Overview of the proposed methodology.

3.2 Mechanical design

The system is modeled as:

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \tag{3.1}$$

3.3 Code

3.3.1 Docker setup

3.3.2 ROS2 controller

The proposed algorithm is shown in Listing **??**.

```python
import numpy as np
from scipy.optimize import minimize

def cost_function(x, *args):
    """
    Compute the cost for given parameters.

    Args:
        x: Parameter vector
        args: Additional arguments

    Returns:
        Cost value
```

```python
    """
    # Extract parameters
    param1, param2 = x[0], x[1]

    # Compute cost
    cost = param1**2 + param2**2

    return cost

# Initial guess
x0 = np.array([1.0, 1.0])

# Constraints
constraints = {'type': 'ineq', 'fun': lambda x: x[0] + x[1] - 1}

# Optimize
result = minimize(cost_function, x0,
                  method='SLSQP',
                  constraints=constraints)

print(f"Optimal solution: {result.x}")
print(f"Final cost: {result.fun}")
```

Listing 3.1: Proposed optimization algorithm

3.4 Experimental Setup

Pictures of real robot. Define what we measure, how we measure. What is concidered working, when does it fail. Experimental testing of friction coefficients, damping... to implement in code

3.4.1 Hardware Configuration

Schematic of robot linked with maxon drivers, software ...

3.4.2 Results

Results-> graphs, values...

Table **??** summarizes the experimental results.

Table 3.1: Comparison of different control strategies.

Method	Rise Time (s)	Settling Time (s)	Overshoot (%)	RMSE
PID Control	2.3	5.1	12.5	0.089
LQR Control	1.8	4.2	8.3	0.065
Proposed Method	1.5	3.8	5.1	0.042

As shown in Table **??**, the proposed method achieves superior performance.

Chapter 4

Results and Discussion

This chapter presents the experimental results and their analysis.

4.1 Experimental results

Figure **??** shows the simulation results.

Figure 4.1: Simulation results showing system response over time.

4.2 Discussion

The results demonstrate that...

4.2.1 Performance Analysis

The improved performance can be attributed to...

4.2.2 Comparison with Existing Methods

Compared to traditional approaches, our method offers:

1. Faster response time
2. Lower overshoot
3. Better disturbance rejection

4.2.3 Limitations

The current approach has some limitations:

- Computational complexity scales as $O(n^2)$
- Requires accurate system model
- Sensitive to measurement noise

Chapter 5

Conclusion and Future Work

This chapter summarizes the contributions and suggests directions for future research.

5.1 Summary

This thesis investigated [problem] and developed [solution].

The main findings are:

- We developed a novel methodology that...
- Experimental results showed that...
- The proposed approach outperforms existing methods by...

5.2 Contributions

The key contributions of this work are:

1. **Theoretical contribution:** A new framework for...
2. **Methodological contribution:** An efficient algorithm for...
3. **Practical contribution:** A prototype implementation demonstrating...

5.3 Future Work

Future research directions include:

- Extension to multi-agent systems
- Integration with machine learning techniques
- Real-time implementation on embedded systems
- Validation on industrial case studies

Appendix A

2D drawings

2d drawings with tolerances

A.1 CNC Part 1

pic

A.2 CNC Part 2

A.3 CNC Part 3

Appendix B

Additional Results

This appendix presents supplementary experimental data.

B.1 Extended Simulation Results

Additional simulation scenarios...

Appendix C

Source Code

Complete source code is available at: https://github.com/username/repository